

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2023-24

Paper: CS-102 — Digital Logics and Circuits

Time: Three Hours

Maximum Marks: 70

*Instructions: Attempt **five** questions including Question No. 1, which is compulsory. Write your Roll No. at the top immediately on receipt of this question paper.*

Question 1.

- (a) How does changing the base of a number system impact the representation and arithmetic operations performed on numbers? [3]
- (b) What is an unweighted and weighted code? Provide an example of each. [3]
- (c) What is the primary difference between combinational and sequential circuits? [3]
- (d) Explain the concept of a multiplexer and its applications. Design a 4-to-1 multiplexer using basic logic gates. [3]
- (e) How many 1's are present in the binary representation of $4 \times 256 + 8 \times 16 + 12$? [2]

Question 2.

- (a) Compute: $(B1.21)_{16} + (211)_4 + (131)_8 = (?)_8$ [4]
- (b) How is a negative number represented in sign-magnitude form? Calculate the sum of -36 and $+52$ using signed number addition. [5]
- (c) Express the following functions in Sum of Minterms and Product of Maxterms and implement using a universal gate of your choice: [5]
 - (i) $F(A, B, C, D) = A'D + C'D' + ABD$
 - (ii) $F(A, B, C) = (ABC + BC)(A'B'C + BC)$

Question 3.

- (a) Minimize the expression with don't-care conditions $F(A, B, C, D) = \sum(0, 1, 5, 13, 14, 15) + \sum d(3, 4, 7, 10, 11)$ using K-map. Draw the logic diagram of the minimized circuit. Why are reflected codes used in K-Map? [7]
- (b) Determine the Prime-Implicants and essential Prime-Implicants of the following Boolean function using the Tabulation method: [7]

$$F(A, B, C, D, E, F, G) = \sum(20, 28, 30, 52, 60)$$

Question 4.

- (a) What are Half and Full-Adder circuits? Explain in detail with a logic diagram and truth table. For the addition of two 8-bit numbers, what is the required number of half adders and full adders? Explain with a logic diagram. [5]
- (b) Design a circuit that compares two 4-bit numbers X and Y . The outcome of the comparison is specified by three binary variables indicating whether $X = Y$, $X > Y$, or $X < Y$. [5]
- (c) Implement a Full-Subtractor circuit using a multiplexer. [4]

Question 5.

- (a) What is ROM? Design a combinational circuit using a ROM that accepts a 2-bit number and generates an output binary number equal to the cube of the input number. [5]
- (b) What is a code converter, and why is it important in digital systems? Design a BCD to Excess-3 code converter. [5]
- (c) What is a Programmable Logic Array (PLA)? Explain its basic structure and functionality. [4]

Question 6.

- (a) Explain the various types of flip-flops used in digital circuits: SR, JK, D, and T flip-flops. Explain how each can be implemented using basic logic gates. For each type, describe its truth table and characteristic equation. [6]
- (b) What is an excitation table for SR, D, T, and JK flip-flops, and how does it differ from a truth table? [4]
- (c) What is a Register? Design a 4-bit Register. [4]

Question 7.

- (a) What are counters and their types? Design a 3-bit synchronous up-counter using a T flip-flop. Explain with a timing diagram. [7]
- (b) What is the primary purpose of state reduction and state assignment in sequential circuit design? Explain with an example. [7]